

Data Mining – Frequent Itemset Generation (Apriori Algorithm)

Given Data Transactions:

TID	ITEMS
1	Bread, Milk, Chips, Mustard
2	Beer, Diaper, Bread, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk, Chips
5	Coke, Bread, Diaper, Milk
6	Beer, Bread, Diaper, Milk, Mustard
7	Coke, Bread, Diaper, Milk

Total Transactions = 7

Therefore, **Minimum Support** = $40\% \times 7 = 2.8 \approx 3$ transactions

1.Frequent 1-Itemsets (L1):

Item	Count	Frequent?
Bread	6	Frequent
Milk	6	Frequent
Diaper	6	Frequent
Beer	4	Frequent
Coke	3	Frequent
Chips	2	Not in frequent
Mustard	2	Not in frequent
Eggs	1	Not in frequent

Therefore, **L1** = {Bread, Milk, Diaper, Beer, Coke}

2. Frequent 2-Itemsets (L2):

Itemset	Count	Frequent?
{Bread, Milk}	5	Frequent
{Bread, Diaper}	5	Frequent
{Milk, Diaper}	5	Frequent
{Beer, Diaper}	4	Frequent
{Beer, Bread}	3	Frequent
{Beer, Milk}	3	Frequent
{Coke, Milk}	3	Frequent
{Coke, Bread}	2	Not in Frequent
{Coke, Diaper}	3	Frequent

L2 includes the following itemset

{Bread, Milk} (4)
{Bread, Diaper} (5)
{Milk, Diaper} (5)
{Beer, Diaper} (4)
{Beer, Bread} (3)
{Coke, Milk} (3)
{Coke, Diaper} (3)
{Beer, Milk} (3)

3.Frequent 3-Itemsets (L3):

Itemset	Support
{Bread, Milk, Diaper}	4
{Beer, Milk, Diaper}	3
{Beer, Bread, Diaper}	3
{Coke, Bread, Diaper}	2 not satisfy
{Coke, Milk, Diaper}	3

L3 Included

{Bread, Milk, Diaper} (4)

{Beer, Bread, Diaper} (3)

{Beer, Milk, Diaper} (3)

{Coke, Milk, Diaper} (3)

4. Frequent 4-Itemsets:

None (no itemset meets minimum support)

Therefore, Using the Apriori Algorithm with minimum support of 40%, the largest frequent itemset size is 3.